

Addis Ababa University

College of Health Sciences

School of Medicine



Integrated Modular Undergraduate Medical Curriculum

.....



2020/21

TABLE OF CONTENTS

- 1. Background**
- 2. Rational**
- 3. Objective**
- 4. Graduate Profile**
- 5. Admission Requirements**
- 6. Educational Strategy**
- 7. Duration of the Program, Curriculum structure and phases**
- 8. Graduation Requirement**
- 9. Degree Nomenclature**
- 10. List of Modules/Courses**
- 11. Sequence of Modules**
- 12. Teaching methodology**
- 13. Assessment of students**
- 14. Grading**
- 15. Program evaluation and quality assurance**
- 16. Requirement for promotion and grading**
- 17. Module description**
- 18. Resource profile**

1. BACKGROUND

Addis Ababa University was founded in 1950, while the faculty of medicine, which was formally approved on Tuesday, March 26th, 1963, was officially established in 1964. Postgraduate training began in 1979. The items considered in the first curriculum, training program, and organization of the Medical faculty (Ethiop Med J 1964) were: full international recognition, clinical training, training in public health and social medicine, training concerning service in the rural areas of Ethiopia, integrated institute of medical sciences, postgraduate training and research.

A later curricular review revealed deficiencies such as lack of integration of the courses and inadequate emphasis on social medicine. A survey among general practitioners revealed, among others, the needs for:

- more practical skill training
- skills in emergency surgical and life-saving clinical procedures.
- basic sciences to be relevant and clinically oriented.
- placing adequate emphasis on the health problem of the community.
- competence in the areas of Leadership, Communication Skills, Ethical Principles, and ICT skills.

In addition, according to the Lancet Commission on medical education report (2010), the problems in the training of health professionals are systemic with a mismatch of competencies to patient and population needs; poor teamwork; narrow technical focus without broader contextual understanding; episodic encounters rather than continuous care; predominant hospital orientation at the expense of primary care; quantitative and qualitative imbalances in the professional labor market; and weak leadership to improve health-system performance. Many others also reflected on the problem and recommended educational reform.

A new trend in the training of health professionals is system-based to improve the performance of health systems by adapting core professional competencies. Thus, in designing the new modular curriculum, the above-mentioned issues were taken into consideration.

2. Rationale

In recent years, there has been an increasing focus on the modular approach of learning in higher education institutions. Modularization is an emerging trend in educational thinking that shifts the traditional method of instruction to an outcome-based learning paradigm. Available data also indicates that the principles of modular-qualifications-based learning outcomes are being adopted in many countries. In addition, modularization of the curriculum enables to maximize student learning and performance

Ethiopia is in the process of rapid change in all spheres, including economic development, the springing up of new industries, new occupations emerging, and specializations and professions constantly arising. The learning process should take advantage of the change and align with work-based competencies through modularization of curricula. The tool that helps to design, describe, and deliver programs and award higher education qualifications is the ECTS system (European Credit Transfer and Accumulation system). Thus, ECTS assists institutions in shifting to a learner centered approach that puts learners' needs and expectations at the front.

There were limited experiences regarding modular curriculum implementation in a few faculties of Addis Ababa University before the Business Process Reengineering (BPR). Currently, almost all of the Masters programs of Addis Ababa University are modularized. Since 2008/09, AAU is implementing a modular curriculum in most fields of studies at undergraduate level. The consensus was reached also at a national level to move towards modularization of the undergraduate medical curriculum. Hence, the first version of this curriculum is developed and presented for implementation in 2014/15 GC.

Based on recommendations on revision workshops conducted in 2017, 2018, and 2019 minor curriculum amendments were made. The current revision, 2019-20 to 2020 -21 academic year, was performed as per the recommendation on Education Roadmap (MoSHE introduced 17 additional courses to all curricula). Accordingly, several workshops: at Addis Ababa, Adama, Bishoftu, and Gondar were conducted engaging stakeholders from MoSHE, MoH, Medical schools, Professional Associations, and partners to integrate the newly added courses into the curriculum.

3. Objectives

General objectives

- To produce competent, compassionate and community oriented doctors for Ethiopia with internationally recognized standard of excellence.

Specific Objectives

- To teach and train medical students to become not only highly competent doctors but also appreciate the value of lifelong learning, to be equipped with good communication skills and to understand the importance of resource management.
- To enable the graduating medical students provide the highest standards of clinical service.
- To train medical doctors who behave according to ethical and legal principles
- To enable the graduates conduct scholarly research in the clinical, biomedical and public health spheres.
To train medical students who will be active in community health activities.

4. Graduate Profile

The graduates of the medical school are expected to possess the following competencies to practice as general medical practitioners

Medical Expert

- Understand the normal and abnormalities in body structure/ function
- Recognize normal and abnormal human behavior.
- Understand the important determinants of health.
- Know the molecular, cellular, biochemical and physiological mechanisms that maintain the body's homeostasis.
- Understand the principles of drug action and its use, and efficacy of various therapies.
- Be able to formulate an appropriate diagnostic hypothesis based on sound, scientific understanding of normal and abnormal structure and function, disease processes, social and psychological context of illness.
- Be able to formulate management plans for individual care of patients, and apply it for treating the common medical, surgical, obstetrics, psychiatric and pediatric illnesses.
- Be able to recognize personal limitations and identify medical and surgical problems which need specialist consultation and referral.
- Have the knowledge of disease prevention and health promotion.
- Be able to obtain from the individual patient the information necessary to formulate and test a diagnostic hypothesis by taking appropriate physical, social and psychological history, physical examination techniques, and using investigations selectively.
- Have an appropriate skill to perform minor operations including safe delivery.
- Have the capacity to investigate and manage epidemics.
- Have the skill of disease prevention and health promotion.
- Be able to handle emergencies: Trauma, Surgical emergencies, Obstetrics emergencies, medical emergencies and pediatrics
- Be able to understand medical literature and accept responsibility to contribute to the advancement of medical knowledge through conducting research.

Communicator

- Communicate effectively and sensitively with patients and their relatives to enable them to undertake decisions as equal partners.
- Recognize that good medical practice depends on a mutual understanding and relationship between the doctor, the patient and the family with respect for patient's welfare, culture, diversity, beliefs and autonomy.
- Be able to communicate effectively with colleagues and other healthcare professionals, the community, and other sectors.
- Be involved in recording, reporting and analyzing health data.
- Be willing to teach and transfer knowledge and skill to colleagues and students.

Collaborator

- Effectively work within a healthcare team to achieve optimal patient care.
- Effectively work with other health professionals to prevent, negotiate, and resolve interprofessional conflict.
- Recognize that good medical practice depends on a mutual understanding and relationship between the doctor and his/her professional team.

Manager

- Participate in activities that contribute to the effectiveness of their healthcare organizations and systems;
- Manage practice and career effectively;
- Allocate finite healthcare resources appropriately;
- Serve in administration and leadership roles, as appropriate.
- Have the capacity to properly manage healthcare institutions.

Health Advocate

- Advance the well-being of individual patients, communities, and populations.
- Identify advocacy activities as important activities for the individual patient, for populations of patients and communities.
- Identify and collaboratively address broad health issues and the determinants of health.

Scholar

- Be willing to take part in continuing professional development to make sure that they maintain high levels of clinical competence and knowledge.
- Be able to gain, assess, apply and integrate new knowledge and have the ability to adapt to changing circumstances throughout their professional life.
- Critically evaluate information and its sources, and apply this appropriately to practice decisions;
- Be able to understand medical literature and accept responsibility to contribute to the advancement of medical knowledge through conducting research.

Professional

- Demonstrate commitment to patients, profession, and society through sound ethical practice;
- Make the care of the patient her/his first concern, treat patients with politeness and consideration, respect their dignity, privacy and point of view without regard to background, culture, language, religion, race and point of view.
- Demonstrate a commitment to their patients, profession, and society through participation in profession-led regulation;

- Be able to recognize the essential elements of the medical profession, including moral and ethical principles and legal responsibilities underlying the profession.

5. Admission Requirements

- The admission to the programme will be based on students' interest **and** her/his academic performance which includes achievements on the EUEE, Freshman 1st semester GPA and Medicine Entrance examination. The Ministry of Education and the university determine the proportion of each examination. Female applicants are encouraged and will be given priority for admission.
- Students must at all times be physically **and** mentally capable of completing the medical courses and subsequently practicing as medical doctors. Applicants with significant physical disabilities such as complete loss of vision, complete loss of hearing, and serious musculoskeletal disabilities incompatible with the routine practice of medicine and severe psychiatric or mental problems that can be detrimental to safe practice may not be accepted by the school of medicine.
- Physical and mental capability shall be determined during a screening procedure performed by a special medical board set up by the Ministry of Education and the school of medicine academic commission during admission.
- Applications of foreign citizens or Ethiopian living abroad holding BSC degrees in related fields of study will be evaluated by AC standing committee and decided by AC. These applicants are expected to pay the tuition fee as foreign students set by Addis Ababa University.

6. Educational strategy

Integrated: The curriculum is designed to teach the modules in an integrated manner both horizontally and vertically.

Early clinical exposure: Students in their pre-clerkship years will receive clinically oriented Lectures by clinicians and will get clinical exposure in the wards to relate the biomedical sciences to clinical medicine.

Community oriented: During the clerkship training, students will have attachments to health centers in selected departments. In addition, during their final year, there will be a community based rural attachment.

Competency based: It is important to make sure that students have acquired the required competency after the completion of each module. This is designed to be confirmed through continuous assessment of students in the learning outcomes.

Hospital based: During the clerkship period, a great proportion of the attachment is based in hospitals. Students will use logbooks to document the skills learned and confirmed to have gained the required competency.

Student centered: Students will be encouraged to practice student centered learning. This is

facilitated by improving the access to ICT.

7. Duration of the program, Curriculum structure & phases

The total duration of training is for 6 & half years

- Premedical phase (Freshman) 1 year
- Preclinical phase 2 years
- Clinical phase 2 and a half years
- Internship 1 year

1. Pre-medical phase (Freshman year)

It shall be offered for 1 year to strengthen the general science and humanities base of the student and to bring the knowledge of students from different high school backgrounds to an equally acceptable level. These shall be considered as basic common courses and will be uniform across the different universities in Ethiopia.

2. The pre-clinical phase

The pre-clinic phase shall be provided for a maximum of two-year. Its main objective is the teaching of biomedical sciences relevant to the making of doctors, with the aim of laying a solid biomedical science foundation in the medical student in preparation for the clinical teaching. As a red thread for future problem-oriented teaching initiatives, integrative strategies and student-centered approaches shall be initiated, encouraged and nurtured. As this curriculum is integrated both vertically and horizontally, clinical lectures and sessions will be incorporated in each preclinical module.

3. The Clinical teaching phase

The clinical teaching (Two and a half years) shall include intensive practical, theoretical and self-directed learning opportunities, and students should have enough exposure to patients in order to acquire knowledge, attitude and skills necessary to assume clinical responsibility upon graduation. In clinical years, more student-centered and problem-oriented teaching methods shall be employed. The spectrum shall include lectures, bed-side teaching, seminars, teaching rounds, clinico-pathological conferences, death rounds, management sessions, side-lab exercises, case portfolio preparation, clinical skill exercising under supervision and others.

The clinical teaching shall be defined as rotating in the following disciplines. The four attachments Surgery, Pediatrics and child health, Internal Medicine, and Gynecology and Obstetrics shall be considered as long term rotations. Rotations to Psychiatry, Ophthalmology, Dentistry, Radiology, Dermatology and ORL (ENT) shall constitute the short term attachments. Communication skills with patients, physical diagnosis, record keeping, medical ethics and professionalism principles shall be provided as a bridging course at the start of the clinical years. As this curriculum is integrated both vertically and horizontally, pre-clinical lectures and sessions will be incorporated in each clinical module.

4. **The public health training**

The public health programme shall use community diagnosis surveys, students' group research, community work, participation in public health services, mini projects design & implementation, etc. as its teaching methodology for acquiring public health skills. It shall be constituted by different modules delivered during the pre-clinical and clinical phases and the actual rural community health program during the final year of clinical training where the medical students travel to a community based practice.

5. **The internship**

It shall be a yearlong learning and service giving exercise that shall be devoted to practical experience. Its aim shall be providing opportunity for the novel doctor for adequate exposure to patients, service setup, actual practicing situations, and updating her/his knowledge and skill.

8. Graduation Requirements (Refer to the detail in section 16)

1. All students should pass successfully in all modules and attachments by obtaining a minimum grade of C+, except for the premedical modules/courses.
2. Successfully complete the 6 & a half years of training and a total ECTS of 403.
 - A minimum cumulative Grade Point Average (CGPA) of 2.0
 - Students should successfully pass the final year qualifying examination.
 - Interns should successfully complete all the internship rotations.

9. Degree Nomenclature

After successful completion of the programme, the graduate shall be awarded a degree named “**Doctor of Medicine**” (MD), “በሕክምና የዶክተርነት ዲግሪ” with all its responsibilities, rights and privileges

10. List of Modules/Courses

1. FLEn 1011 Communicative English Language and Skills I
2. Phys 1011 General Physics
3. Psyc 1011 General Psychology

4. Math 1011 Mathematics for Natural Science
5. LoCT 1011 Critical Thinking
6. SpSc 1011 Physical Fitness
7. GeES 1011 Geography of Ethiopia and the Horn
8. FLEn 1012 Communicative English Language and Skills II
9. Anth 1012 Social Anthropology
10. EmTe 1012 Introduction to Emerging Technology
11. Econ 1012 Economics
12. Chem 1012 Organic Chemistry
13. Incl 1012 Inclusiveness
14. GLOT 1012 Global trend
15. MCIE 1012 Moral and Civics Education
16. Anth 1012 Social Anthropology
17. EmTe 1012 Introduction to Emerging Technology
18. Econ 1011 Economics
19. Chem 1012 Organic Chemistry
20. SNIE 1012 Inclusiveness
21. GITr 1012 Global trend
22. BSOF 1011 Body structure, organization and functions - I
23. BSOF 1012 Body structure, organization and functions - II
24. MHMG 1021 Metabolic Homeostasis and Molecular Genetics
25. BCDT 1031 Basic concepts of disease and therapy
26. PUBH 1012 Measurement of Health and Diseases
27. NMBC 1051 Neoplasia and molecular basis of cancer
28. BLIM 1041 Blood and immunity
29. MICF 1061 Musculoskeletal, integumentary and craniofacial regions
30. PUBH 2022 Basics of Biostatistics and Epidemiology; Research Methodology
31. CAPU 2071 Cardiopulmonary System
32. NMDG 2081 Nutrition, Metabolic diseases and GIT
33. USTX 2091 Urinary system and Toxicology
34. ENRP 2051 Endocrinology and Reproduction
35. NSBE 2111 Neuroscience and Behavior
36. INFD 2121 Infectious Diseases
37. CMCS 3012 Clinical Methods & Communication skills
38. MELP 3012 Medical Ethics, Law and Professionalism I
39. IMED 3011 Internal Medicine I
40. SURG 3011 Surgery I
41. OBGY 3011 Obstetrics and Gynecology I
42. PUBH 4012 Health Promotion and Disease Prevention

43. PUBH 4022 Health Management and Policy
44. ORLY 4012 Otorhinolaryngology (ENT)
45. OPTH 4012 Ophthalmology
46. DENT 4012 Dentistry
47. DERM 4012 Dermatology
48. ITGN 4014 Introduction to Gender
49. MGMT 1012 Entrepreneurship
50. PAED 4021 Pediatrics and Child Health II
51. OBGY 4021 Obstetrics and Gynecology II
52. DIRA 4012 Diagnostic Radiology
53. MELP 3012 Medical Ethics, Law & Professionalism II
54. PUBH 4052 Rural community Health Practice
55. IMED 4021 Internal Medicine II
56. SURG 4021 Surgery II
57. PSYC 4012 Psychiatry
58. EMRM 4012 Emergency Medicine
59. IMED 5031 Internal Medicine
60. OBGY 5031 Obstetrics & Gynecology
61. PAED 5031 Pediatrics & Child Health
62. SURG 5031 Surgery
63. ANES 5031 Anesthesia
64. EMRM 5031 Emergency Medicine
65. PSYC 5031 Psychiatry

11. Sequence of Modules

I. Freshman Courses (1 Year)

Semester I

S.N	Course Code	Course Name	ECTS
1.	FLEn 1011	Communicative English Language and Skills I	5
2.	Phys 1011	General Physics	5
3.	Psyc 1011	General Psychology	5
4.	Math 1011	Mathematics for Natural Science	5
5.	LoCT 1011	Critical Thinking	5
6.	SpSc 1011	Physical Fitness	P/F
7.	GeES1011	Geography of Ethiopia and the Horn	5
			30

Medicine Entrance Exam

Semester II (Pre-Med)

S.N	Course Code	Course Name	ECTS	Parallel
1.	FLEn 1012	Communicative English Language and Skills II	5	
2.	Anth 1012	Social Anthropology	4	
3.	EmTe 1012	Introduction to Emerging Technology	5	
4.	Econ 1011	Economics	5	
5.	Chem 1012	Organic Chemistry	5	
6.	SNIE 1012	Inclusiveness	4	
7.	GITr 1012	Global trend	4	
8.	MCIE 1012	Moral and Civics Education	4	
		Total	36	

II. Pre-clinical Year I (Med Year 1)**Semester I**

S.N	Module number	Module	ECTS
1	BSOF 1011	Body structure, organization and functions - I	6
2	BSOF 1012	Body structure, organization and functions - II	6
3	MHMG 1021	Metabolic Homeostasis and Molecular Genetics	8
4	BCDT 1031	Basic concepts of disease and therapy	6
		Total	26

Vacation: 2 weeks

Semester II

S.N	Module number	Module	ECTS
1	PUBH 1012	Measurement of Health and Diseases	10
2	NMBC 1051	Neoplasia and molecular basis of cancer	6
3	BLIM 1041	Blood and immunity	5
4	MICF 1061	Musculoskeletal, integumentary and craniofacial regions	10
		Total	31

Vacation: 1 month

End of year 1.

III. Pre-Clinical Year II (Med Year 2)

Semester I

S.N	Course number	Course	Total ECTS
1	PUBH 2022	Basics of Biostatistics and Epidemiology; Research Methodology	6
2	CAPU 2071	Cardiopulmonary System	13
3	NMDG 2081	Nutrition, Metabolic diseases and GIT	8
4	USTX 2091	Urinary system and Toxicology	5
		Total	32

Vacation: 2 weeks

Semester II

S.N	Course number	Course	Total ECTS
1	ENRP 2051	Endocrinology and Reproduction	13
2	NSBE 2111	Neuroscience and Behavior	10
3	INFD 2121	Infectious Diseases	6
		Total	29

Exit Exam A: at the end of year 2

Vacation: One month

End of year 2

IV. Clinical Year I (Med Year 3)

S.N	Module number	Module	ECTS
1	CMCS 3012	Clinical Methods & Communication skills	4
2	MELP 3012	Medical Ethics, Law and Professionalism I	4
3	IMED 3011	Internal Medicine I	14
4	SURG 3011	Surgery I	14
5	OBGY 3011	Obstetrics and Gynecology I	14
6	PAED 3011	Paediatrics and Child Health I	14
		Total	64

End attachment examinations: at the end of each attachment.

Vacation: Mid-year vacation (after two long attachments) is 2 weeks while end year vacation is 1 month.

End of year 3

V. Clinical year II (Med Year 4 and 5)

Semester I

1	PUBH 4012	Health Promotion and Disease Prevention	5	
2	PUBH 4022	Health Management and Policy	3	
3	ORLY 4012	Otorhinolaryngology (ENT)	4	
4	OPTH 4012	Ophthalmology	4	
5	DENT 4012	Dentistry	4	
6	DERM 4012	Dermatology	4	
7	ITGN 4014	Introduction to Gender	2	Parallel
8	MGMT 1012	Entrepreneurship	3	
		Total	29	

Vacation: 2 weeks

Semester II

S.N	Module number	Course	ECTS
1	PAED 4021	Pediatrics and Child Health II	8
2	OBGY 4021	Obstetrics and Gynecology II	8
3	DIRA 4012	Diagnostic Radiology	4
4	MELP 3012	Medical Ethics, Law & Professionalism II	4
5	PUBH 4052	Rural community Health Practice	8
		Total	32

End attachment Examinations: at the end of attachments.

Vacations: 2 weeks

Semester III

S.N	Course number	Course	Total ECTS
1	IMED 4021	Internal Medicine II	8
2	SURG 4021	Surgery II	8

3	PSYC 4012	Psychiatry	8
4	EMRM 4012	Emergency Medicine	8
		Total	32

A final qualifying examination in Surgery, Internal Medicine, Pediatrics & Child Health, Gynecology-Obstetrics and Public Health will be given at the end of Clinical year 2.

Revision time: for final qualifying exam is 3 weeks while the exam period is 1 month.

End of Year 4: Vacation: is 1 month

VI. Internship: (Med Year 6)

- Internal Medicine (10 weeks)
- Obstetrics - Gynecology (10 weeks)
- Pediatrics and Child Health (10weeks)
- Surgery (10 Weeks)
- Anesthesia (2 weeks)
- Emergency Medicine (5 weeks)
- Psychiatry (3 weeks)

Interns will have a one week long vacation during each longer attachment.

S.N	Course number	Course	Total ECTS
1	IMED 5031	Internal Medicine	12
2	OBGY 5031	Obstetrics & Gynecology	12
3	PAED 5031	Pediatrics & Child Health	12
4	SURG 5031	Surgery	12
5	ANES 5031	Anesthesia	2
4	EMRM 5031	Emergency Medicine	7
7	PSYC 5031	Psychiatry	3
		Total	60

12. Teaching Methodology

The following main effective methods of teaching will be utilized.

- Interactive lectures
- Laboratory demonstrations and practice

- Seminars
- Simulated practice (clinical skills lab)
- Clinical attachments
 - i. Rounds
 - ii. Bed-sides
 - iii. Case discussions
 - iv. Operation theatre teaching
- Community-based learning
- Mentoring
- Individual and group research
- Whole group session

13. Assessment

The final grade of the students in each module shall be computed based on a fair balance of the progressive assessment, final written examination, oral examination, and practical examination. The format by which the completion of each module is assessed will be as depicted in the syllabus. The status determination shall be at the end of each semester in the premedical (Freshman) and preclinical phases and at the end of each year during the clinical phase.

Progress from the preclinical to clinical phase will be determined by internal evaluation of each module and qualification examination given at the end of year 2 that shall be known as Exit Exam A (Pre-clinical Qualifying Examination). There shall be a guideline in place on the implementation of this qualification examination. As per the guideline, a student who fails the exit exam will take a remedial exam. If a student fails the remedial exam he/she shall take the exam next year with the following batch. If a student does not pass the exam for the 3rd time he/she will be dismissed and recommended to be transferred to other fields of study in the University.

Progress from Clinical I to Clinical II will be determined by internal evaluation at the end of each attachment.

Final year medical students' progress to the internship phase in the five long term attachments (Internal medicine, Surgery, Pediatrics, Obstetrics/Gynecology and Public health) shall be determined by internal examinations (end attachment examinations) and a qualification examination known as Exit Exam B (Clinical Qualifying Examination). The qualification examination shall consist of a written and clinical examination.

The clinical examinations shall be conducted in the presence of an external examiner as much as possible. The role of the external examiner, besides evaluating the students, is to give opinions on the suitability of the student evaluation process and the teaching programs, taking into account the views of teachers and students and teaching facilities. Assessment of students in the other departments will be at the end of each attachment.

Interns shall be assessed by their clinical knowledge, conduct, professionalism, and attitude towards the patient, the profession and the healthcare team as Excellent (A), Very good (B), Good (C), Unsatisfactory, or Fail (D or F). Interns who fail in internship shall repeat the rotations for a minimum of 10 weeks.

Attendance is mandatory for medical students. As stipulated in the Senate legislation, medical students are required to fulfill a 100% attendance in all teaching activities. Failure to achieve this may result in an automatic repeat of the course, module, or clinical attachment.

14. Grading

Letter grades shall be given based on the points earned out of 100 and it has a fixed scale as described in the table below (adopted but modified from the AAU senate legislation of 2019)

Raw Mark Interval [100 %]	Corresponding fixed Number Grade	Corresponding Letter Grade	Status Description
90 - 100	4.0	A ⁺	Excellent
85 - 90	4.0	A	
80 - 84	3.75	A ⁻	
75 - 79	3.5	B ⁺	Very Good
70 - 74	3.0	B	
65 - 69	2.5	C ⁺	Good
60 - 64	2.0	C	Satisfactory
50 - 59	1.0	D	Unsatisfactory
< 50	0	F	Fail

15. Programme evaluation and quality assurance.

This curriculum is open to periodic evaluation. Proper monitoring and evaluation tools shall be developed by **the associate dean office** of the school of medicine, Medical Education Unit, quality assurance mechanisms.

16. Requirement for promotion and graduation

1. “Remedial Examination” shall mean an examination that is offered to a student in good academic standing (CGPA $\geq$ 2.0), but who, according to the grading system of the school of medicine, obtained “D” and/or “F” in modules other than the premedical modules in order to remove these deficiencies or an “F” in the premedical modules.

2. For all courses in this curriculum, except the premedical modules, the passing grade shall be “C”. “D” may be accepted in the premedical modules.
3. A student shall be allowed to take remedial examination only once. A student is allowed to repeat a module only once during the program.
4. A student who fails to achieve CGPA of 2.0 but has $\text{CGPA} \geq 1.75$ after remedial examination shall repeat the module which the student has failed. If the student failed for the module he/she repeated shall be dismissed for good.
5. A student who achieves CGPA between 1.5 and 1.75 at the end of the year shall repeat the whole year.
6. Any student who fails to score CGPA of 1.5 at the end of the year shall be dismissed with no chance of a remedial or repeating the year.
7. **The following rules shall apply to students during the premedical courses.**
 - a. In the premedical phase, a student can be allowed to progress to the pre-clinical year I and take a remedial examination in modules where he/she has scored a “D” or F” **only if** the semester GPA ≥ 1.5
 - b. Otherwise, he/she will be dismissed and re-admitted the following year to take the courses again. The student, by no means, shall be allowed to enter Pre-clinical year II without removing the failing grades during Pre-clinical I. A pre-medical module with a “D” grade may be recorded as it is during graduation if the student has a good academic standing.
8. **The following rules shall apply to PC I and PC II students.**
 - a. At the end of each semester, if a student fails in one module, the student shall be allowed to register for the coming semester and a remedial examination should be given **within four weeks** of the beginning of the following semester. If the student fails to pass the remedial examination, he/she shall repeat the module during the following year, hence repeats the whole year. If a student failed for the module repeated he/she shall be dismissed for good.
 - b. If a student fails in two modules in a semester, he/she shall be allowed to take remedial examination only if the student has a good academic standing in that semester. Otherwise, he/she shall repeat the module in the coming year, hence repeats the whole

year and be admitted with the next batch of students. The student will not register for the coming semester

- c. Failure in three modules results in automatic dismissal. The student shall be readmitted and repeat the modules during the following year. If a student fails in one of the module, he/she shall be dismissed for good.

9. The following rules shall apply for Clinical year I students.

- a. Students must score a minimum of “C” in all rotations to enter Clinical II.
- b. Grades shall be determined by progressive assessment, end attachment clinical examination and written examination.
- c. Students who score D in only one major attachment shall be allowed to progress to CN II and will take remedial examination only once provided their academic standing permits. The remedial examination shall be given within two weeks of commencement of the next year and the “D” grade shall be removed.
- d. Students who fail in the remedial examination shall repeat the whole attachment. They shall not be allowed to continue with the same batch. If a student failed after reattachment exam he/she will be dismissed.
- e. Students who score D in two or more rotations or students who score an F in one or more major attachments shall repeat the rotation/s. They shall be admitted with the next batch.

10. Requirements to sit for the final qualifying examination

- a. Students must have a minimum CGPA of 2.0 and above.
- b. They must complete all premedical, preclinical, clinical and public health modules with a minimum grade of “C”.
- c. The registrar should present the list of prospective candidates who will sit for the final qualifying examination for approval by the school academic commission.

11. The following rules shall apply for Year IV and V students sitting for the final qualifying examination.

- a. Candidates must score a minimum of “C” in all the five qualifying examinations in order to enter their year of Internship.

- b. The end attachment results of the final year medical students shall not be recorded in the grade report.
- c. The overall year IV examination grades shall be composed of
 - i. The year IV end attachment clinical examination: 40%
 - ii. Final written examination: 30%
 - iii. External qualification examination: 30%
- d. The above results shall be recorded in the grade reports against the course title and credit hours of year IV.
- e. Candidates who fail in one or two qualification examinations shall repeat each rotation for 3 months and should pass an internal examination.
- f. Candidates who fail in three or more examinations shall repeat the whole year. They will repeat each attachment and shall pass the qualifying examinations. They will not be allowed to graduate with the same batch.
- g. Candidates who failed in qualifying examination in public health shall repeat a public health attachment for six weeks.
- h. Students who fail to score a minimum of “C” in the short attachments shall be allowed to take remedial examination.
- i. If a student failed in two short attachments he/she shall take remedial exam before qualification
- j. If a student failed in 3 or more, he/she shall re attaches the modules.

17. Module description

FRESHMAN YEAR SEMESTER II (PRE-MEDICAL PHASE)

Module title: Communicative English Language and skills II
Module code: FLEN 1012
Module credit (ECTS): 5
Course pre-requisite: Communicative English Language and Skills I
Placement: Freshman year Semester II
Module description:

Module title: Social Anthropology (Medical Sociology and Anthropology)
Module code: Anth 1012
Module credit (ECTS): 4
Placement: Freshman year Semester II
Module description:

Module title: Introduction to Emerging Technology in Medicine

Module code: EmTe 1012

Module credit (ECTS): 5

Placement: Freshman year Semester II

Module description:

Module title: Economics

Module code: Econ 1012

Module credit (ECTS): 3

Placement: Freshman year Semester II

Module description:

Module title: Organic Chemistry

Module code: Chem 1012

Module credit (ECTS): 5

Placement: Freshman year Semester II

Module description:

Module title: Inclusiveness

Module code: Incl 1012

Module credit (ECTS): 2

Placement: Freshman year Semester II

Module description:

Module title: Global trend

Module code: GLOT 1012

Module credit (ECTS): 2

Placement: Freshman year Semester II

Module description:

Module title: Moral and Civics Education

Module code: MCIE 1012

Module credit (ECTS): 4

Placement: Freshman year Semester II

Module description:

PRECLINICAL MODULES

At the end of preclinical modules learners are expected to: Understand and gain knowledge, skills

of clinically oriented and integrated biomedical sciences; develop self-learning skills; develop holistic understanding of the body, diseases, and pharmacological aspects of therapeutics; understand the biology of microorganisms related to human diseases; develop evidence based learning and understand preventive, curative, promotive and rehabilitative aspects of health.

1. Body Structure, organization and function (Module I & II)

Module Code: BSOF 1011 and BSOF 1012
Module credit (ECTS): 12
Placement: Year I, Semester I
Prerequisite module: Pre-medical courses

Module Description:

This module is intended to describe body structure and function; and also molecular mechanisms that operate in the living cell. It also deals with developmental periods with emphasis on the beginning of human development, formation of embryonic discs and organogenesis, the fetal period, human birth defects, formation of all body systems and molecular regulation of development.

2. Metabolic Homeostasis and Molecular Genetics (Module III)

Module Code: MHMG 1021
Module credit (ECTS): 8
Placement: Year I, Semester I
Prerequisites: Module I & II

Module description:

This module describes the basic principles of enzyme catalysis, regulation and clinical significance of enzymes as biomarkers; biosynthesis, action mechanism of major hormones in metabolic signal transduction pathway. Metabolism and regulation of carbohydrates, lipids, proteins as well as the principles of bioenergetics, cellular energy generation will be introduced. The importance of genetic information storage, transcription and translation and application of DNA Technology in clinical medicine, proteomic and metabolomics will be described.

3. Basic Concepts of Disease and Therapy (Module IV)

Module Code: BCDT 1031
Module credit (ECTS): 6
Placement: Year I, Semester I
Prerequisites: Module I-III

Module description:

This module is designed to prepare a student to be a competent Medical Doctor. Upon completion of the module student will be able to describe the general principles and introduction of pathology, Microbiology, Parasitology, biochemistry and Pharmacology including principles and causes of disease, classification of microbes, structures of microbes, microbial virulence factors host-parasite relationship, principles of sterilization and disinfection and principles of pharmacokinetics and pharmacodynamics.

4. Measurement of Health and Disease (Module V)

Module code: PUBH 1012
Module credits (ECTS): 10
Placement: Year I, Semester II

Unit I: Epidemiology

Description: The module covers definitions, principles and concepts, estimating the occurrence of disease, interpreting the occurrence of disease by person, place, and time; validity and reliability and concepts of disease causation and determinants. Students are also introduced to the design and conduct of epidemiologic studies (descriptive, analytical and experimental), data quality and principles of screening and surveillance, clinical epidemiology, and the control of selected health problems.

Unit 2: Biostatistics**Description**

This module introduces theoretical and practical aspects of introductory biostatistics which is relevant to undergraduate medical training program. Specifically it deals with measurement scales, presentation of scientific data, calculation of central tendency measures, basic probability, sampling techniques, tests of significance, hypothesis testing, correlation and regression, vital statistics and introductory demography.

5. Neoplasia and molecular basis of cancer (Module VI)

Module Code: NMBC 1041
Module credit (ECTS): 6
Placement: Year I Semester II
Prerequisites: Module I-IV

Module description: This unit of learning offers a comprehensive presentation of cancer as a disease. It will consider the genetic abnormalities that contribute to cancer development, mechanisms of mutagenesis and DNA repair. The molecular mechanisms that lead from a normal cell to a cancer cell. The role of oncogenes, protein products of oncogenes, cancer suppressor genes, protein products of tumor suppressor genes, genes that regulate apoptosis, the multi hit mechanisms of cancer and topics related to biology of tumor growth. The morphological, biochemical, molecular characteristics that distinguish normal cells from cancer cells. Basics morphologic and behavioral characteristics of neoplasia.

6. Blood and Immunity (Module VII)**Module Code:** BLIM 1051**Module credit (ECTS):** 5**Placement:** Year I, Semester II**Prerequisites:** Module I-IV & VI**Module description**

The module describes the structure and function of blood and its constituents; immune system and structure of lymphoid tissue and organs; disorders of blood and immunity and treatment. It also includes inflammation and treatment of inflammatory diseases as well as tissue repair.

7. Musculoskeletal, Integumentary Systems, and Craniofacial (Module VIII)**Module code:** MICF 1061**Module credit (ECTS):** 10**Placement:** Year I, Semester II**Prerequisites:** Module I-IV & VI-VII**Module Description:**

This module describes the development, macroscopic and microscopic structures, molecular mechanisms and function, as well as pathological and pharmacological aspects of the musculoskeletal and integumentary systems.

8. Determinants of Health and Disease & research Methodology (Module IX)**Module Code:** PUBH 2022**Module credit (ECTS):** 6**Placement:** Year II, Semester I**Pre-requisite:** Module V**Unit I: Environmental Health****Description:**

The module includes two parts: **an introduction to human ecology-Environment** that deals with various issues of interactions between man and his immediate environment; and **Hygiene, Sanitation, and Environmental Health** with particular emphasis to Ethiopian condition. The module focuses on the ecology of human activities affecting the environment and that environment that poses adverse health effect to his health and surrounding. Specific areas include interactions with water, air, culture, physical chemical factors, population and developmental activities; fundamentals of environmental health on water, food, human waste, vectors of public health importance, basics of radiational hygiene, housing, and waste management.

Unit II: Behavioral Health Sciences

Description:

The undergraduate level Behavioral Health Sciences course at the School of Public Health, College of Health Sciences at Addis Ababa University introduces the concepts of behavioural health sciences with special emphasis on health promotion and health education based on theoretical principles and international conventions. It also covers behavioral, social and environmental health determinants, communication, education and educational methods, predicting behavior using theoretical models, evaluation of behavioral change and related concepts and principles from research perspectives. The course mainly focuses in the techniques of planning, implementation, monitoring and evaluation of health education programs, qualitative research and essential behavioral changes and communication skills for health practitioners.

Unit 3: Public health Nutrition**Description:**

This module describes the main principles of applied human nutrition and their application to identify and analyze food and nutrition related problems at the population level; effective strategies and programs to address nutritional issues, various dimensions of the problem of food and nutrition security and the corresponding intervention strategies in developing countries, focusing on the situation in Ethiopia.

Research Methodology

This module includes the principal notions and components of research for undergraduate medical students at the school of medicine. It deals with the ways how to define research questions, formulate problem statement, develop a research project proposal, how to collect research data and analyze and interpret data.

9. Cardiopulmonary (Module X)

Module Code: CAPU 2071

Module credits (ECTS): 13

Placement: Year II, Semester I

Prerequisites: Module VI - VIII

Module description:

This module deals with normal structural organization and developmental changes, as well as, cardiac muscle energy production, biochemical basis for maintenance of cardiovascular and pulmonary hemostasis. It also deals with the different developmental, structural and functional disorders of cardiovascular and pulmonary systems together with their treatments.

10. Nutrition, metabolic diseases and gastrointestinal tract (Module XI)

Module Code: NMGI 2081

Module credits (ECTS): 8

Placement: Year II, Semester I

Prerequisites: Module VI – VIII and X

Module description:

This module covers the gross anatomy of abdomen and its viscera; it also deals with microscopic and developmental anatomy of digestive and excretory systems. Moreover, it describes the digestive and kidney functions with acid base balance. Furthermore, it describes basic principles of human nutrition, metabolism and nutrient requirement under different physiological status and major nutritional deficiency diseases together with their etiology and epidemiological aspects relevant to clinical evaluations. It also deals with different diseases involving the digestive and excretory system together with pharmacological control of disturbance.

11. Excretion and Toxicology (Module XII)

Module Code: EXTX 2091
Module credits (ECTS): 5
Placement: Year II, Semester I
Prerequisites: Module VI – VIII and X, XI

Module description:

This course describes the regional anatomy of the posterior abdomen, as well as kidney histology, embryology and function in regards to acid-base balance. Also addressed are the different diseases involving the excretory system, toxicokinetics, toxicodynamics and the management of specific toxicants. Finally, the microbiology/parasitology of urinary tract infections will be studied, and students will be taught practical skills for the identification of uropathogens from clinical specimens.

12. Endocrinology and reproduction (Module XIII)

Module Code: ENRP 2101
Module credits (ECTS): 13
Placement: Year II Semester II
Prerequisites: Module VI – VIII and X-XII

Module description

This module describes gross, microscopic and developmental anatomies and physiological function of endocrine and reproductive systems. It also deals with the introduction of infectious disease and therapy of endocrine and reproductive disorders

13. Neuroscience and behavior Module XIV)

Module Code: NSBE 2111
Module credits (ECTS): 10
Placement: Year II Semester II
Prerequisites: Module VI – VIII and X-XIII

Module description:

This module describes the development, macroscopic, and microscopic structures, functions, neurotransmitters metabolism, pathological alterations, infections, pharmacological modulation of the nervous system.

14. Infectious diseases (Module XV)**Module Code:** INFD2121**Module credits (ECTS):** 6**Placement:** Year II Semester II**Prerequisites:** Module VI-VIII, X-XIV

Module description:

The module focuses on the conveying of essential knowledge, skills and attitude in acquisitions of competencies in tropical diseases caused by bacteria, viruses, fungi, and parasites in disease pathology, laboratory skills, epidemiology, treatment and disease control methods.

CLINICAL YEAR MODULES

1. Clinical Methods and Communication Skills (Bridging course)**Module Code:** CMCS 3012**ECTS:** 4**Prerequisite:** Successful completion of biomedical sciences**Placement:** Clinical year 1 Semester I

Module description:

The aim of the module is to introduce the student to the knowledge, attitude, and skills required to communicate with a patient effectively and elicit the health problems, explain and demonstrate common manifestations of diseases and their correlation with the underlying pathophysiological mechanism and introduce the institutional organization of patient care and appropriate documentation of information. The subject will be delivered in the form of lectures, ward rounds and bedside teaching. Student assessment will be done using progressive assessment, written and practical examination.

2. Medical Ethics, Law and Professionalism I**Module code:** MELP 3012**ECTS:** 4**Prerequisites:** Completion of pre-clinical training**Placement:** Clinical year I Semester I

Module description:

This course is designed to introduce students to ethical issues in health care policy, planning, and service delivery in the global context. Through the application to practical decision making at all levels of the health care system of a broad range of ethical principles and moral theories, the students will gain an understanding of 1) the definition of health care ethics and bioethics, including comparative interpretations; 2) the complexities of health care ethics and bioethics in the global context; 3) the protocol and systems in place to ensure adherence to ethical principles; and 4) their specific roles and responsibilities to ensure such adherence.

3. Surgery I

Module code: SURG 3011
ECTS: 14
Prerequisites: Completion of pre-clinical training and the physical diagnosis module
Placement: Clinical year I

Module description:

The module deals with taking relevant history and performing examinations to reach a clinical diagnosis of common surgical problems. The student will describe the etiology, pathophysiology, clinical features, diagnosis and principles of treatment of common surgical diseases in Ethiopia and Africa with emphasis to surgical emergencies. The course is given in the form of systematic lectures, seminars, tutorials, bedside teaching, clinical teaching inwards, operating rooms, and outpatient clinics. Students are attached to orthopaedics for 2 weeks during this period. Student performance at the end of the course is assessed through continuous assessment in seminars, participation in tutorials, bedside teachings and teaching rounds, and written and practical clinical examinations.

4. Surgery II

Module code: SURG 4021
ECTS: 8
Prerequisite: SURG 3011
Placement: Clinical year II

Module description:

The course deals with an introduction to operative surgery and training in the performance of minor and emergency operations. The course is aimed at consolidating the knowledge and skills acquired during SURG I, focusing on the management of common surgical diseases in a district hospital setting. acquiring essential surgical skills to enable the medical graduate to handle simple emergency surgical procedures independently; and recognizing complex surgical problems that require tertiary level care. There is a 1-week rotation in orthopaedics during this course. Learning-teaching methods include tutorials, seminars, bedside teachings, teaching in ward rounds, operating rooms and OPD, and a one-week workshop on essential surgical skills using clinical simulators. Student assessment is based on continuous assessment, clinical examination and oral examination

5. Internal medicine I

Module number: IMED 3011
ECTS: 14
Prerequisite: Basic medical sciences and physical diagnosis course.
Placement: Clinical year I

Module description:

The overall goal of the Internal Medicine course is to help students understand and obtain practical experience in the recognition, evaluation, diagnosis and management of adult patients with medical illnesses. The module is designed to prepare graduate medical doctors who are qualified and competent to identify, diagnose and provide holistic management as well as preventive measures for medical diseases in adults and adolescents. The module is an eleven week clinical experience. The clinical rotation is designed to develop patient-related cognitive and decision-making skills. The staff physician for each service directs patient responsibilities, reading assignments, and educational activities. Students are encouraged to observe, evaluate and participate in discussions of patient care. The student will learn about various diseases with emphasis on prevalent diseases in Ethiopia. The subject will be delivered in the form of lectures, clinical teachings, students' grand-round and medical circus. Assessment of students will be done using progressive assessment, written and practical examination.

6. Internal medicine II

Module number: IMED 4021
ECTS: 8
Prerequisite: Internal Medicine I
Placement: Clinical year II

Module description:

The objectives of this module are to enable the student manage common disorders including acute emergencies, identify difficult cases in order to refer them to proper health care facility for further evaluation and treatment. This second rotation (Internal Medicine II) is a six weeks clinical experience. The clinical rotations are designed to develop patient-related cognitive and decision-making skills. The staff physician for each service directs patient responsibilities, reading assignments, and educational activities. Students are encouraged to observe, evaluate and participate in discussions of patient care. The module will be delivered in the form of Tutorials, standard case discussions, clinical teachings, student grand-round medical circus and management sessions. Assessment will be done using progressive assessment, written and practical examinations. External examiners will also be involved in the evaluation of students before their graduation

7. OBSTETRICS AND GYNECOLOGY I

Module number: GYOB 3011
ECTS: 14
Prerequisite: Biomedical sciences
Placement: Clinical year I

Module description:

The aim of the course is to enable the student to the basic principles of obstetrics and gynecology and make them understand the normal physiology and anatomy of a patient, evaluation of a female patient-proper history taking and physical examination and appreciate and manage common clinical problems.

The course will include fundamentals of obstetrics and gynecology like endocrine changes and diagnosis and follow up of normal pregnancy, conduct of normal labor and postnatal checkup. Introduction to operative obstetrics (forceps, vacuum, destructive delivery, episiotomy and cesarean section etc.) will be provided.

The course is delivered through lectures, seminars, tutorials, and bedside teaching, skills laboratory and OR exposure. Assessment is composed of progressive assessment, written and practical examination.

8. OBSTETRICS AND GYNECOLOGY II

Module number: OBGY 4021
ECTS: 8
Prerequisite: Obstetrics and Gynecology I
Placement: Clinical year 2

Module description:

The aim of the course is to enable the student to acquire the required level of competence in knowledge, skill and attitude basic to the clinical and public health practice of obstetrics and gynecology. The course will enable the students to acquire knowledge, skill and confidence to diagnose and propose management outline of a particular disease using additional measures like laboratory facilities and other diagnostic tools.

The course is delivered through lectures, seminars, tutorials, and bedside teaching, skills laboratory and OR exposure. Assessment is composed of progressive assessment, written and practical examination.

9. PAEDIATRICS AND CHILD HEALTH I

Module number: PAED3011
ECTS: 14
Prerequisite: Completion of the biomedical courses
Placement: Clinical year 1

Module description:

The aim of the module is to enable the student to collect relevant history, perform complete physical examination and select relevant laboratory investigations to reach at the diagnosis of the child's problems.

The course presents to the students to the basic principles of child health problems by imparting knowledge about basic mechanisms of selected diseases peculiar to pediatrics, growth and development, and principles of management. The various diseases of pediatrics are taught with more emphasis on prevalent diseases in Ethiopia. The students will also be assisted to develop appropriate attitudes towards self-learning and prevention of common health problems. The course

is offered through series of lectures, seminars, bedside teaching and preceptorship, Students' assessment includes written examinations, clinical examinations, viva voce and progressive assessment.

10. Paediatrics and Child Health II

Module number: PAED 4021
ECTS: 8
Prerequisite: Paediatrics and child health I
Placement: Clinical year II

Module description:

The aim of the module is to enable the student to manage common child health problems, emergencies and practice preventive paediatrics. The module deals with prescription writing, principles of management of common illnesses and emergencies, health supervision, maternal and child health and ambulatory services. The module is delivered through seminars, clerkship, grand rounds, and supervised practice in community health facilities. Student assessment is composed of progressive assessment, written and clinical examinations and viva voce. Qualifying assessments involve external examiners.

11. Dentistry

Module number: DENT 4012
ECTS: 4
Placement: Clinical year 2

Module description:

The objective is to enable the student to manage common diseases of the teeth and oral cavity. The module deals with the anatomy of the teeth and organs and tissues of the oral cavity. It also addresses diseases of the teeth and oral cavity, tooth extraction and anesthesia for dental manipulation. The module is delivered by way of lectures, seminars and clinical practice. Evaluation is based on progressive assessment.

12. Dermatovenereology

Module number: DERM 4012
ECTS: 4
Placement: Clinical year 2

Module description:

The objectives of the module are to give additional knowledge on tropical dermatology, to introduce dermatological diseases in greater depth, to enable them to manage common dermatological disorders and to identify difficult cases in order to recommend (refer) to the appropriate health care facility. The module includes common dermatological problems prevalent in Ethiopia. The subject will be delivered in the form of lectures and clinical teaching. Evaluation will be done using progressive assessment, written and practical examinations.

13. Diagnostic radiology**Module number:** DIRA 4012**ECTS:** 4**Placement:** Clinical year II

The purpose of the module is for the student to have basic knowledge on how to read simple radiographs and to know the principles and applications of modern imaging modalities focusing on emergency ultrasound applications. The module will include radiological anatomy, basic radiology physics; procedures (contrast gastrointestinal tract studies, principles of ultrasound and fluoroscopic images). The subject will be administered through lectures, tutorials and case discussions. Students will be evaluated through a reporting session and written examination.

14. Medical Ethics, Law and Professionalism II**Module code:** MELP 4012**ECTS:** 4**Prerequisites:** MELP 3012**Placement:** Clinical year II Semester III

Module description:

This course is the second part of the medical ethics course provided in year III. Through the application to practical decision making at all levels of the health care system of a broad range of ethical principles and moral theories, the students will gain an understanding and practical application of the ethical stems described in the previous module.

15. Ophthalmology**Module number:** OPTH 4012**ECTS:** 4**Placement:** Clinical year II

Module description:

The module is designed to set up undergraduate medical students competent enough in identifying and managing common ophthalmic problems encountered in the community. Up on completion of the course, students will be able to diagnose and medically treat common ophthalmic problems. In addition, they will be able to identify those patients who would require referral to an ophthalmologist. The course presents to the students the basic principles of ocular and periocular problems by imparting knowledge about the anatomy and physiology of the eye and vision, basic mechanisms of selected ophthalmic and neuro-ophthalmic diseases, and principles of management. The students will also be assisted to develop appropriate attitudes towards the importance of vision with all the qualities for proper execution of day to day activities. This can be achieved through periodic ophthalmic checkup especially in association with systemic illnesses. A positive attitude towards early intervention of most of the ophthalmic problems is also imparted to students through the clinical demonstrations. In addition to this, the course will assist students to develop the necessary skills to apply the acquired knowledge in clinical situations. The course is offered through a series of lectures, seminars, and clinical demonstrations. The assessment will

be based on progressive assessment and written examinations.

16. Otorhinolaryngology (ENT)

Module number: ORLY 4012

ECTS: 4

Placement: Clinical year II

Module description:

The purpose of this module is for the senior medical students is to impart knowledge, skills and attitudes pertaining to the diseases of the organ systems the specialty deals with: common congenital, inflammatory, degenerative, metabolic, traumatic and neoplastic affections of the ear, nose, throat, head and neck organs. The course familiarizes the trainee to basic equipments and facilitates acquisition of skills to identify normal individual variations, enable to examine patients and recognize problems, understand basic/ routine/ tests, do minor procedures, prioritize patients' management and plan timely referrals.

17. Emergency medicine

Module number: EMRM 4012

ECTS: 8

Placement: Clinical year II

Module description:

At the end the student will be able to understand emergency medical problems and manage them and will develop competency on life saving techniques. The student will also have knowledge and skills on emergency medical services organization and leadership. It deals with emergency medical problems with emphasis on life saving procedures. In the course threatening emergencies in medicine, paediatrics, surgery, obstetrics and other areas will be dealt. Students will have attachments to emergency rooms, Anesthesiology and critical care and will have special demonstration time on life saving surgical skills. The teaching will be conducted through lectures case discussions, video demonstrations and simulations and skills laboratory. Evaluation will be through written exam viva and progressive assessment

18. Psychiatry

Module number: PSYC 4012

ECTS: 8

Placement: Clinical year II

Module description:

The main purpose of this module is to provide students with knowledge of main psychiatric disorders, the principles underlying modern psychiatric theory, commonly used treatments and their side effects, and provide a basis on which to continue to develop this knowledge. In addition to this, the module will assist students to develop the necessary skills to apply the acquired knowledge in clinical situations. This module is designed to encourage students to develop the appropriate attitudes necessary to respond empathically to psychological distress in all medical

settings. The course covers descriptive psychopathology, classification, methods of eliciting symptoms, formation of syndrome (s) and reaching to a class diagnosis using decision-tree. Psychosocial development, dynamics of social interaction, interview techniques and cultural aspects of psychiatric disorders are dealt with during this early phase. The course is delivered through case demonstrations and case-taking by the students, case discussions and seminars. Early management of psychiatric cases in an integrated general hospital/health centre/clinic is emphasized. Assessment is by written examination.

19. Health promotion and Disease prevention

Reproductive health, Public health nutrition

Module number: PUBH 4012

ECTS: 5

Prerequisite module code(s): Basic concepts of Epidemiology, Biostatistics, Health Management and Behavioral Health Sciences

Module description:

This module provides the concepts of reproductive health, organization and management of family planning programs, different maternal Health services, measurement of maternal morbidity and mortality, causes of maternal health problems, interventions to improve maternal morbidity and mortality and deals with causes and determinants of child and adolescent health problems and interventions. This module also describes the main principles of applied human nutrition and their application.

20. Health Services Management and Policy

Module number: PUBH 4022

ECTS: 3

Prerequisite module code(s): None

Description of the Module

The module provides medical students with basic knowledge of the principles of health management and introduces concepts of health economics; it discusses paradigms and contextual emphasis the health policies of Ethiopia; it also sensitizes students with major contemporary global and country specific health strategies and initiatives including Primary Health Care and Millennium Development Goals. The module focuses on familiarizing medical students to health systems including structures and functions and their roles in managing and leading health care organizations. Moreover, it enables students enhance their skills in teamwork, familiarization to health policies and strategies. Writing reports and making presentations are some of the skills that are embedded in the module in line with student centered approach.

21. Module title: Rural Community Health Team Training Practice

Module number: PUBH 4052

ECTS: 8

Prerequisites: Modules on the introduction of public health, epidemiology and biostatistics, health service management and reproductive health and nutrition are prerequisite for this module

Description of the module:

This module has contained two courses. In the first course students will learn health research methods and in the second module, they will be attached to rural community health activities. Immediately after the completion of the first course, students will be engaged in rural community health activities. This module will be delivered to medical students during the fourth year second semester of the medical curriculum. It introduces the principal notions and components of research for undergraduate medical students at the School of Medicine. It deals with the ways how to define research questions, formulate problem statement, develop a research project proposal, how to collect research data and analyze and interpret data.

The rural attachment is also designed to apply research knowledge and offer an integrated series of learning experiences which will sensitize medical students to important health problems and provide the students with the conceptual and analytic tools needed to critically examine the health needs of a defined population group; develop programmes to meet these needs; and evaluate the impact of these programmes on the health status of the population

INTERNSHIP

Module number: SURG5031, IMED5031, PAED5031, OBGY5031, ANES5031, EMRM5031, PSYC5031

ECTS: 60

Prerequisites: Successful completion of all clinical year courses

Duration: 12 months

The attachment deals with the assignment of the candidate in rotating clinical attachments. The aim is to expose the candidate to the real life situation working under supervision. Interns will rotate in Surgery, Internal medicine, Pediatrics, Obstetrics and gynecology, Anesthesia, Emergency medicine and Psychiatry. Assessment will be based on progressive assessment.

18. Resource profile

18.1 Human resources:

Rank	Female	Male	Total
Professor	1	25	26
Associate Professor	11	74	85
Assistant Professor	103	202	305
Lecturer	43	83	126

Assistant Lecturer	9	6	15
Technical Assistant/Lab Technician	5	6	11
Total	184	411	568

18.2. Other resources:

- A 800-bed capacity hospital
- 7 affiliated hospitals and 10 health centers
- Community training and research sites: Butajera and Zeway Campus
- A new and spacious Library building accommodating all library services
- Two computer labs with more than 350 functioning computers
- Easily accessible fast wireless internet connection
- Medical Simulation laboratory
- Surgical skill laboratory
- Six biomedical sciences laboratories including 2 dissection halls